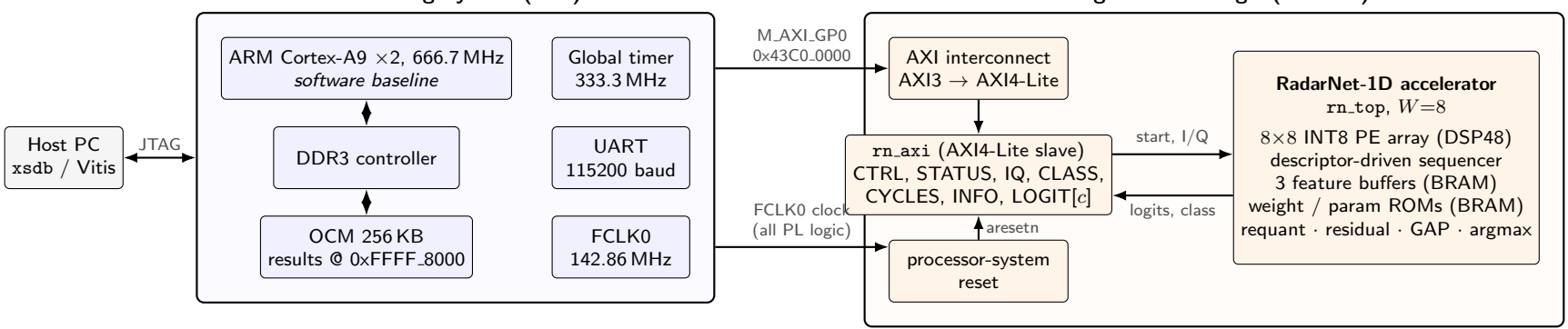

Processing System (PS7)
Programmable Logic (xc7z020)
Host PC
xsdb / Vitis
JTAG
ARM Cortex-A9 ×2, 666.7 MHz
software baseline
Global timer
333.3 MHz
DDR3 controller
UART
115200 baud
OCM 256 KB
results @ 0xFFFF_8000
FCLK0
142.86 MHz
M_AXI_GP0
0x43C0_0000
FCLK0 clock
(all PL logic)
AXI interconnect
AXI3 → AXI4-Lite
rn_axi (AXI4-Lite slave)
CTRL, STATUS, IQ, CLASS,
CYCLES, INFO, LOGIT[c]
aresetn
processor-system
reset
start, I/Q
logits, class
RadarNet-1D accelerator
rn_top, W=8
8×8 INT8 PE array (DSP48)
descriptor-driven sequencer
3 feature buffers (BRAM)
weight / param ROMs (BRAM)
requant · residual · GAP · argmax